

Theorem 1. *Assume $\text{char}(k) = 0$ and put $R := k[\varepsilon]/(\varepsilon^2)$. There exists a finite-dimensional R -algebra $\mathcal{A}$ which has a faithful R -free module of infinite rank, but has no faithful R -free module of finite rank.*

Proof. Let $S := k[e, f]/(e, f)^2$ and let V have basis $\{u_n, v_n\}_{n \geq 0}$ with $eu_n = v_n$, $fu_n = v_{n+1}$, and $ev_n = fv_n = 0$; write $E, F \in \text{End}_k(V)$ for the actions (this representation is faithful). Define $A, C \in \text{End}_k(V)$ by $A(u_n) = 0$, $A(v_n) = (-2n - 1)u_n$, and $C(u_n) = C(v_0) = 0$, $C(v_n) = 2n u_{n-1}$ ($n \geq 1$). Then

$$\text{Id}_V = [A, E] + [C, F]. \quad (1)$$

Let $I := S + SAS + SCS \subseteq \text{End}_k(V)$ and let $\iota : S \hookrightarrow I$ be the inclusion; then I is finite-dimensional and (1) implies $\iota(1) = A \cdot e - e \cdot A + C \cdot f - f \cdot C$ in I . Hence for any finite-dimensional left S -module $W \neq 0$ with action ρ_W , there is no S - S -bimodule map $\Phi : I \rightarrow \text{End}_k(W)$ extending ρ_W , since otherwise Id_W would be a sum of commutators and $\text{tr}(\text{Id}_W) = 0$.

Put $M := V \otimes_k R$; using $R \cong k \oplus k\varepsilon$ we identify $\text{End}_R(M) \cong \text{End}_k(V) \otimes_k R \cong \text{End}_k(V) \oplus \text{End}_k(V)\varepsilon$. Define

$$\mathcal{A} := \{s + \varepsilon T \mid s \in S, T \in I\} \subseteq \text{End}_R(M).$$

Then $\dim_k \mathcal{A} < \infty$, εI is square-zero, and $\mathcal{A}$ acts faithfully on M .

If N were a faithful R -free $\mathcal{A}$ -module of finite rank and $W := N/\varepsilon N$, multiplication by ε gives an isomorphism $\theta : W \xrightarrow{\sim} \varepsilon N$, and

$$\Psi(T)(\bar{n}) := \theta^{-1}(((\varepsilon T) \cdot n))$$

defines an S - S -bimodule map $\Psi : I \rightarrow \text{End}_k(W)$ with $\Psi \circ \iota = \rho_W$, contradiction. $\square$